



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to SRI MADHURA AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

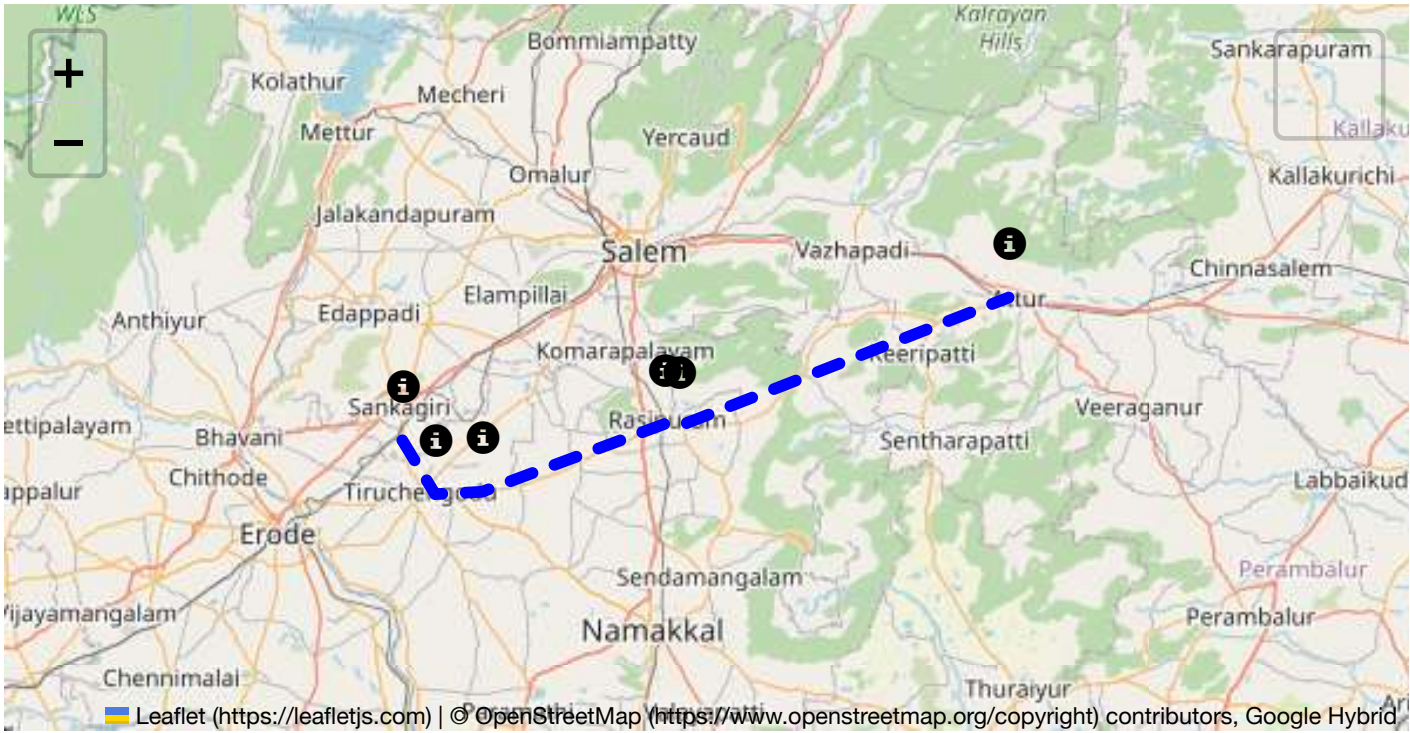
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 99.53 km**  
**Estimated Duration: 2.3 hours**  
**Adjusted Duration (Heavy Vehicle): 2.8 hours**

Start: (11.4381, 77.8734)  
End: (11.601548, 78.586494)



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route begins in Sangagiri, Tamil Nadu, and proceeds through Namakkal, Kavundampalaiyam, Rasipuram, and Koneripatti, finally reaching Vinayagapuram in Attur, Tamil Nadu. Covering about 99.53 kilometers, this route primarily utilizes state highways SH 79 and local roads.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. The monsoon season (October to December) can lead to heavy rains, causing poor visibility and slippery roads. During summer months (March to May), high temperatures might contribute to road surface deterioration and possible heating issues for vehicles.

### 3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7:00 AM - 9:00 AM) and evening rush hours (5:00 PM - 8:00 PM) observe the highest traffic.
- **Congestion-Prone Areas:**
  - Near Namakkal due to a mix of local and through traffic.
  - Rasipuram town can become congested, especially near marketplaces.

### 4. Assessment of Road Quality and Infrastructure

- **State Highways (SH 79):** Generally in good condition but can have patches requiring maintenance.
- **Local Roads:** May have uneven surfaces and limited signage, particularly in rural stretches.

## 5. Suggestions for Alternative Routes for Emergencies

In case of emergencies or road blockages:

- Consider using the Namakkal - Rasipuram - Attur route via SH 95, though longer, offers relatively consistent traffic flow and infrastructure.
- Local directions may offer temporary detours around congestion points.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- Require adherence to specific state regulations for hazardous material transport.
- Restrictions on movement in densely populated areas during certain times.
- Mandatory permits and documentation are required for transportation.

## 7. Overview of Historical Incidents

- Past incidents mainly involve accidents due to weather or road surface issues rather than systemic route hazards.
- Notable incidents include overturned vehicles in slippery conditions during monsoons.

## 8. Environmental Considerations and Sensitive Areas

- Presence of agricultural lands and local ecosystems should be respected; minimize leaks and spills.
- Avoid transport activities through sensitive zones during high rainfall to reduce environmental impact.

## 9. Analysis of Communication Coverage

- Generally good mobile network coverage along highways.
- Possible dead zones in remote stretches around Kavundampalaiyam and Koneripatti, advise carrying a communication alternative like satellite phones or two-way radios.

## 10. Estimated Emergency Response Times

- **Urban Areas (Namakkal and Rasipuram):** Approximately 30-45 minutes depending on traffic.
- **Rural Stretches:** Could extend to an hour or more due to accessibility issues.

## 12. Overall Summary of Risk Assessment

This route poses moderate risks primarily due to environmental factors during monsoon and peak traffic periods. While infrastructure along state highways supports heavy vehicle movement, attention should be given to less maintained local roads. Regulations around hazardous materials necessitate careful planning. Consider alternatives for reliable communication and emergency preparedness in rural areas.

For truck drivers, awareness of high-risk areas, adherence to traffic regulations, and preparedness for adverse weather are crucial for a safe journey. Regular maintenance checks on vehicles carrying hazardous materials can significantly enhance safety.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates            | Speed Limit |
|----|------------|------------|------------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480     | 15 KM/Hr    |
| 1  | U-Turn     | High       | 11.4535263, 78.1964113 | 10 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345     | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44019, 77.87547     | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.39685, 77.88861     | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.38376, 77.89914     | 30 KM/Hr    |
| 6  | Turn       | High       | 11.38186, 77.90258     | 15 KM/Hr    |
| 7  | Blind Spot | Blind Spot | 11.37841, 77.90140     | 10 KM/Hr    |
| 8  | Turn       | Medium     | 11.36754, 77.93919     | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.41354, 78.08075     | 30 KM/Hr    |
| 10 | Turn       | High       | 11.41447, 78.08063     | 15 KM/Hr    |
| 11 | Turn       | High       | 11.41475, 78.08087     | 15 KM/Hr    |
| 12 | Turn       | High       | 11.42964, 78.11587     | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.42447, 78.15264     | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.42611, 78.15532     | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.42651, 78.15557     | 30 KM/Hr    |
| 16 | Turn       | High       | 11.44521, 78.15833     | 15 KM/Hr    |
| 17 | Turn       | Medium     | 11.45566, 78.17361     | 30 KM/Hr    |
| 18 | Turn       | High       | 11.45587, 78.18057     | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.45760, 78.18140     | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.45990, 78.18413     | 30 KM/Hr    |
| 21 | Blind Spot | Blind Spot | 11.46035, 78.18534     | 10 KM/Hr    |
| 22 | Turn       | Medium     | 11.45929, 78.18544     | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.45895, 78.18572     | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.45764, 78.19208     | 30 KM/Hr    |
| 25 | Blind Spot | Blind Spot | 11.45100, 78.20385     | 10 KM/Hr    |
| 26 | Turn       | High       | 11.45084, 78.20378     | 15 KM/Hr    |
| 27 | Blind Spot | Blind Spot | 11.45354, 78.19642     | 10 KM/Hr    |
| 28 | Turn       | Medium     | 11.45117, 78.20024     | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.44993, 78.22152     | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.44951, 78.22263     | 30 KM/Hr    |
| 31 | Turn       | High       | 11.44797, 78.22314     | 15 KM/Hr    |
| 32 | Turn       | Medium     | 11.44756, 78.22414     | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.45353, 78.23584     | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.49617, 78.32335     | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 35 | Turn      | Medium     | 11.49835, 78.32417 | 30 KM/Hr    |
| 36 | Turn      | Medium     | 11.50506, 78.33087 | 30 KM/Hr    |
| 37 | Turn      | Medium     | 11.50865, 78.33966 | 30 KM/Hr    |
| 38 | Turn      | Medium     | 11.51183, 78.34416 | 30 KM/Hr    |
| 39 | Turn      | Medium     | 11.51870, 78.34816 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 11.54842, 78.35627 | 30 KM/Hr    |
| 41 | Turn      | Medium     | 11.56719, 78.37813 | 30 KM/Hr    |
| 42 | Turn      | Medium     | 11.56940, 78.39856 | 30 KM/Hr    |
| 43 | Turn      | Medium     | 11.57829, 78.51245 | 30 KM/Hr    |
| 44 | Turn      | Medium     | 11.58679, 78.54553 | 30 KM/Hr    |
| 45 | Turn      | Medium     | 11.59108, 78.55637 | 30 KM/Hr    |
| 46 | Turn      | Medium     | 11.59253, 78.55778 | 30 KM/Hr    |
| 47 | Turn      | High       | 11.59744, 78.57347 | 15 KM/Hr    |
| 48 | Turn      | High       | 11.59553, 78.57461 | 15 KM/Hr    |
| 49 | Turn      | High       | 11.59538, 78.57598 | 15 KM/Hr    |
| 50 | Turn      | High       | 11.59892, 78.57708 | 15 KM/Hr    |
| 51 | Turn      | Medium     | 11.60109, 78.58219 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                                      | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital          | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode           | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | Jayaa Hospital                            | 11.3848223, 77.9021886 | 30 km/h     | Medium     |
| 14 | hospital | Sudha Hospital                            | 11.4555773, 78.1760125 | 30 km/h     | Medium     |
| 15 | hospital | Government Hospital Blood Bank, Rasipuram | 11.4612082, 78.1849249 | 30 km/h     | Medium     |
| 16 | hospital | Government Hospital, Rasipuram            | 11.4612987, 78.1851369 | 30 km/h     | Medium     |
| 17 | hospital | M.R. Hospital, Namakkal                   | 11.4613462, 78.1861702 | 30 km/h     | Medium     |
| 18 | hospital | dr.thangavel dch hospital                 | 11.4619955, 78.1849181 | 30 km/h     | Medium     |

|    | type     | name                                | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------|------------------------|-------------|------------|
| 19 | police   | Police Station Rasipuram            | 11.4617825, 78.1847411 | 30 km/h     | Medium     |
| 20 | hospital | surya hospital                      | 11.4622163, 78.1840544 | 30 km/h     | Medium     |
| 21 | hospital | Rasi Hospital                       | 11.4618368, 78.1855695 | 30 km/h     | Medium     |
| 22 | hospital | R K Eye Care Center                 | 11.4589079, 78.1913596 | 30 km/h     | Medium     |
| 23 | hospital | Manickkam Hospital                  | 11.4568944, 78.1926061 | 30 km/h     | Medium     |
| 24 | hospital | Snathi Hospital                     | 11.4564245, 78.1932485 | 30 km/h     | Medium     |
| 26 | hospital | Sugam Hospital,Rasipuram            | 11.4553007, 78.1936221 | 30 km/h     | Medium     |
| 27 | hospital | Government Hospital, Namagiripettai | 11.4569484, 78.265263  | 30 km/h     | Medium     |
| 28 | clinic   | Dr. Saravanakumar Clinic            | 11.5842002, 78.4441223 | 30 km/h     | Medium     |
| 29 | hospital | Government Hospital                 | 11.572472, 78.497407   | 30 km/h     | Medium     |
| 31 | hospital | Abi Hospital                        | 11.59319, 78.560129    | 30 km/h     | Medium     |

### Crowded Spots

|    | type        | name                                            | coordinates            | speed_limit | risk_level |
|----|-------------|-------------------------------------------------|------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                    | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                   | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை             | 11.3833608, 77.8970145 | 30 km/h     | Medium     |
| 6  | marketplace | Farmer Market                                   | 11.4146849, 78.0805463 | 30 km/h     | Medium     |
| 7  | school      | Government Higher Secondary School              | 11.4150045, 78.0838172 | 30 km/h     | Medium     |
| 8  | school      | Government school                               | 11.4248027, 78.1323902 | 30 km/h     | Medium     |
| 9  | college     | Paavai Arts And Science College For Women       | 11.4262443, 78.1546987 | 30 km/h     | Medium     |
| 10 | school      | SRV EXCEL Matriculation Higher Secondary School | 11.4299679, 78.152616  | 30 km/h     | Medium     |

|    | type    | name                                    | coordinates               | speed_limit | risk_level |
|----|---------|-----------------------------------------|---------------------------|-------------|------------|
| 11 | college | Thiruvalluvar Govt arts college         | 11.4449352,<br>78.1595562 | 30 km/h     | Medium     |
| 12 | school  | Venkateswara School                     | 11.4468258,<br>78.1595761 | 30 km/h     | Medium     |
| 13 | school  | Venkateswara Middle school              | 11.4472063,<br>78.1596191 | 30 km/h     | Medium     |
| 25 | school  | Govt. Hr. Sec. School                   | 11.4585324,<br>78.1933666 | 30 km/h     | Medium     |
| 30 | school  | Rasi Matric HR Sec School               | 11.5760027,<br>78.5062808 | 30 km/h     | Medium     |
| 32 | school  | Government Boys Higher Secondary School | 11.598743,<br>78.5885472  | 30 km/h     | Medium     |
| 33 | school  | Government Boys Higher Secondary School | 11.599149,<br>78.5888931  | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.38395, 77.89480



**Risk Type: U-Turn**  
**Risk Level: High**  
**Speed Limit: 10 KM/Hr**  
**Coordinates: 11.4535263, 78.1964113**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44019, 77.87547**



**Risk Type: Turn**  
**Risk Level: Medium**  
**Speed Limit: 30 KM/Hr**  
**Coordinates: 11.39685, 77.88861**



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.38376, 77.89914



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.38186, 77.90258



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.37841, 77.90140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.36754, 77.93919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.41354, 78.08075



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41447, 78.08063



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41475, 78.08087



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.42964, 78.11587



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42447, 78.15264



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42611, 78.15532



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42651, 78.15557



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44521, 78.15833



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45566, 78.17361



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45587, 78.18057



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45760, 78.18140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45990, 78.18413



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.46035, 78.18534



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45929, 78.18544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45895, 78.18572



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45764, 78.19208



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45100, 78.20385



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45084, 78.20378



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45354, 78.19642



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45117, 78.20024



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44993, 78.22152



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44951, 78.22263



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44797, 78.22314



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44756, 78.22414



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45353, 78.23584



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49617, 78.32335



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49835, 78.32417



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50506, 78.33087



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50865, 78.33966



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.51183, 78.34416



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.51870, 78.34816



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.54842, 78.35627



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56719, 78.37813



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56940, 78.39856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57829, 78.51245



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58679, 78.54553



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.59744, 78.57347



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.59553, 78.57461



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.59538, 78.57598



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.59892, 78.57708



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.60109, 78.58219

---